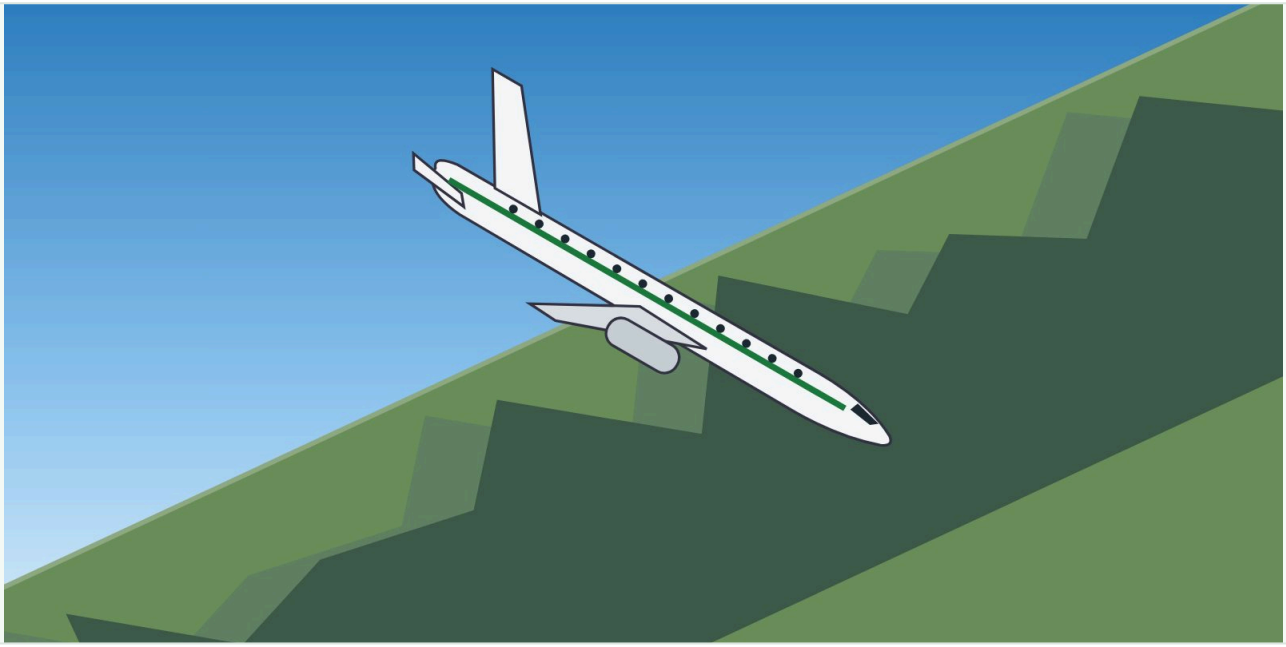
MODULE 6 · RECOVERY

Lesson 22: Nose-Low and High-Bank Upset Recovery

Lesson 22 of 24 · about 8 min

LEARNING OBJECTIVES

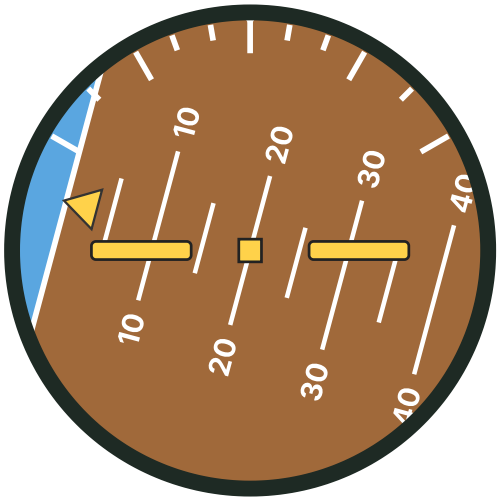
- Apply the generic nose-low recovery technique.
- Explain why the wings must be levelled before pulling.
- Manage speed and g in a high-speed, nose-low recovery.



Nose-low: roll wings level first, then pull.

The situation

The pitch is unintentionally low — for example more than 10° nose-down — often combined with a high bank angle. The speed is usually increasing. The danger is **overspeed**, **structural overload** and ground contact — and, if the aircraft is stalled, a continued uncontrolled descent.



Nose-low: 20° down, 75° right bank

Nose-low, high-bank upset: roll towards the sky pointer before pulling.

Generic technique (based on industry guidance)

1. **Recognise and confirm** the situation. Announce “Nose low”, with the direction of bank.
2. **Autopilot and autothrust — disconnect.**
3. **Recover from the stall, if stalled** — even with the nose low, the wing may be stalled (high descent rate, stall warning, buffet). Reduce the angle of attack first.
4. **Roll in the shortest direction to wings level.** Full aileron and spoiler may be needed. It may be necessary to **unload** (reduce back pressure) to improve roll response.
5. **Recover to level flight:** apply nose-up elevator smoothly, use stabiliser trim if necessary, and adjust thrust and drag as required — reduce thrust and extend the speedbrakes if the speed is high.

Always follow the manufacturer's procedure for your type.

Why roll before pull

- With high bank, pulling back mainly **tightens the turn** and increases the descent — it does not raise the nose. This is how a graveyard spiral develops.
- A **rolling pull** — rolling and pulling at the same time — produces asymmetric structural loads with lower limits.
- Once the wings are near level, the lift vector points up, and the pull raises the nose efficiently.

Bank beyond 90°

If the aircraft is beyond 90° of bank, the shortest way to wings level may require continuing the roll through the inverted position — follow the sky pointer. Unload while rolling. Pulling while inverted only steepens the dive.

Managing speed and g

- Reduce thrust and use speedbrakes as needed to control the speed.
- Pull smoothly to recover — firmly enough to stop the descent, but without exceeding the g limit or triggering the stall warning. Your FCTM gives type-specific guidance.
- Avoid a secondary nose-high upset: as the nose passes the horizon, establish level flight and a suitable speed.

KEY TAKEAWAYS

- Nose-low: automation off → stall recovery if stalled → roll wings level (shortest direction) → then pull.
- Never pull with high bank — it tightens the spiral and risks a rolling pull.
- Control speed with thrust and speedbrakes; pull smoothly within g limits.

KNOWLEDGE CHECK

In a nose-low upset with 80° of bank and increasing speed, what should you do before pulling back?

A. Add thrust

B. Roll wings level in the shortest direction (unloading if needed)

C. Extend the flaps

D. Apply full rudder

Answer: B. With high bank, pulling only tightens the turn. Roll wings level first, then pull smoothly to recover.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.